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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-127

1. Topic (12 words max)

AI-Driven Intelligent Railway Operations System for Sri Lanka

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Railway transportation in Sri Lanka serves millions of passengers daily and remains one of the most affordable and environmentally sustainable transport modes. However, the system continues to face persistent operational challenges, including inaccurate demand estimation, unpredictable delays, inefficient seat allocation, manual fraud detection, and static train scheduling practices. These issues directly affect passengers, inspectors, operators, and policymakers, resulting in overcrowding, revenue leakage, passenger dissatisfaction, and inefficient utilization of limited railway resources.

From a social and economic perspective, unreliable railway services disproportionately affect daily commuters, students, and low-income groups who rely on trains for essential mobility. Technically, the absence of predictive and optimization-based decision support systems leads to reactive rather than proactive operational management. Manual ticket inspection practices further increase labor costs while failing to systematically target high-risk fraud behaviors. Environmentally, inefficient train utilization increases energy consumption per passenger, indirectly affecting sustainability goals.

This research aligns with the **FoC Research Cluster on Smart Mobility and Intelligent Systems** and targets the **Transportation and Public Infrastructure industry vertical**. The project contributes to **UN Sustainable Development Goals (SDGs)**, particularly **SDG 9 (Industry, Innovation and Infrastructure)**, **SDG 11 (Sustainable Cities and Communities)**, and **SDG 13 (Climate Action)** by improving railway efficiency, accessibility, and sustainability through intelligent, data-driven solutions.

Despite global advances in railway analytics, Sri Lanka Railways lacks an integrated, AI-enabled decision support framework that jointly addresses demand forecasting, delay prediction, fraud detection, and schedule optimization. This gap motivates the proposed research.

References

[1] J. G. de Gooijer and R. J. Hyndman, “25 years of time series forecasting,” *Int. J. Forecast.*, vol. 22, no. 3, pp. 443–473, 2006.
[2] S. Bandara et al., “Passenger demand analysis in Sri Lanka Railways,” *Transp. Res. Procedia*, vol. 48, pp. 1234–1245, 2020.
[3] M. Ghofrani et al., “Big data analytics in railway transportation systems,” *Transp. Res. Part C*, vol. 90, pp. 226–246, 2018.
[4] A. Dal Pozzolo et al., “Adversarial drift detection in fraud,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 10, pp. 1–14, 2017.
[5] R. Corman et al., “Railway traffic management in complex networks,” *J. Rail Transp. Plan. Manag.*, vol. 4, no. 2, pp. 1–13, 2014.

6. Brief description of Existing Research and Systems (300 words max)

Existing railway systems in Sri Lanka primarily rely on static timetables, rule-based seat allocation, and manual ticket inspections. Passenger information systems typically provide deterministic arrival times without accounting for uncertainty or delay propagation. Similar limitations are observed in many developing railway networks.

From a research perspective, prior studies on passenger demand forecasting largely focus on short-term or station-level prediction using traditional time-series models such as ARIMA, with limited integration into operational decision-making [1]. Delay prediction research often ignores uncertainty quantification and treats delays as isolated events rather than network-propagated phenomena [3]. Ticket fraud detection studies are commonly framed as supervised classification problems, assuming the availability of balanced and fully labeled datasets, which is unrealistic in real-world railway environments [4]. Furthermore, most scheduling approaches optimize a single objective, such as delay minimization, without offering transparent trade-off analysis among conflicting operational goals [5].

The major gaps identified include:

- Lack of **integrated forecasting and optimization frameworks**
- Absence of **probabilistic and uncertainty-aware delay prediction**
- Limited application of **unsupervised and explainable fraud detection**
- No **Pareto-based multi-objective scheduling** tailored to Sri Lanka Railways

These limitations highlight the need for a unified, intelligent railway decision support framework.

References

[1] J. G. de Gooijer and R. J. Hyndman, "25 years of time series forecasting," *Int. J. Forecast.*, vol. 22, no. 3, pp. 443–473, 2006.
[2] S. Bandara et al., "Passenger demand analysis in Sri Lanka Railways," *Transp. Res. Procedia*, vol. 48, pp. 1234–1245, 2020.
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7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is an **AI-driven, modular decision support framework** designed to enhance operational efficiency, passenger experience, and revenue protection in Sri Lanka Railways.

System Perspective

Inputs

Historical and simulated ticketing data
GPS and station timestamp data
Timetable and rolling stock data
Calendar events and seasonal indicators

Processing

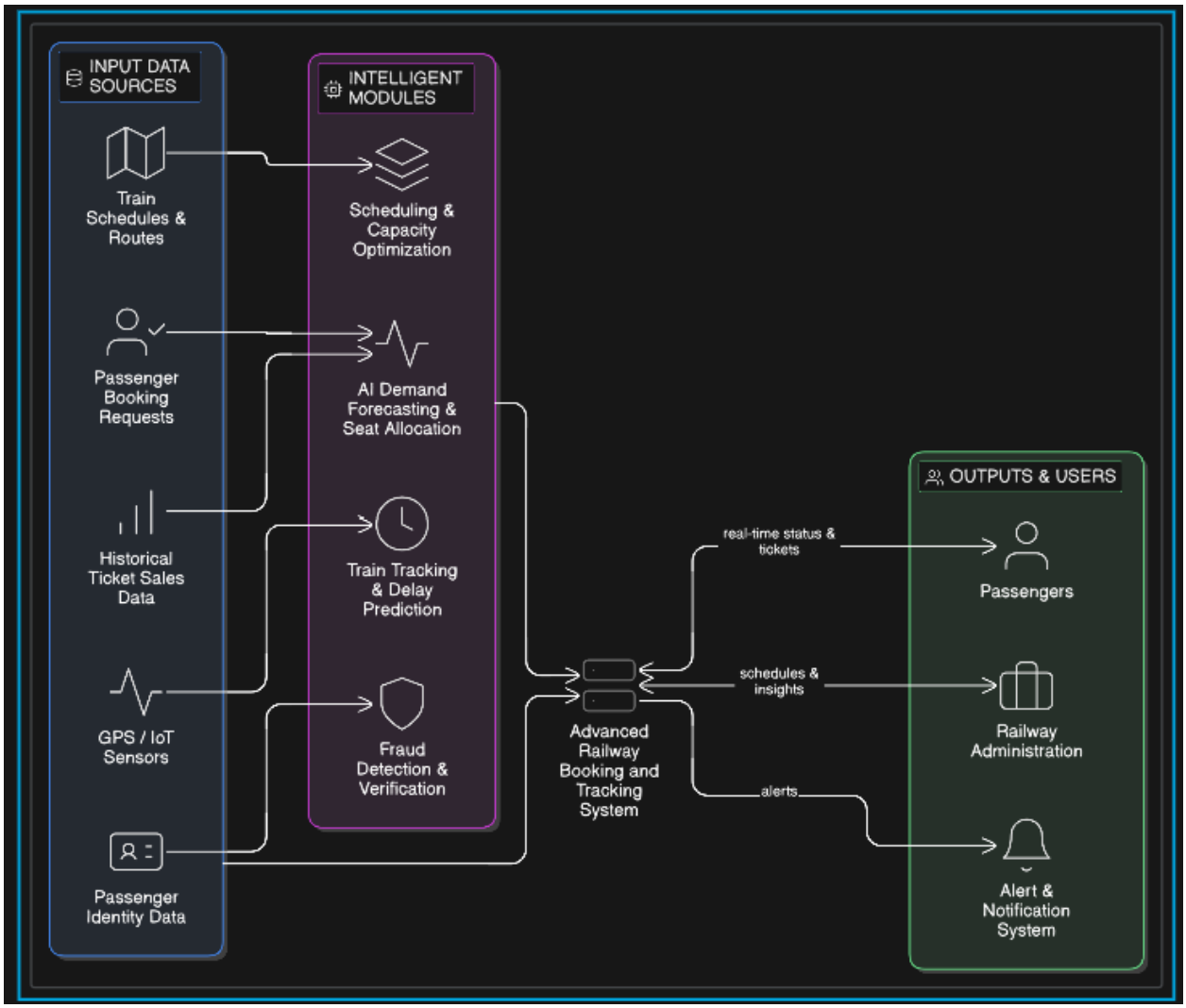
Spatio-Temporal Demand Forecasting Module
Predicts route-level and OD passenger demand using deep learning models.
Adaptive Seat Allocation Module
Allocates seats dynamically using fairness-aware optimization.
Probabilistic Delay Prediction Module
Generates real-time ETAs with confidence intervals using sequence models.
Intelligent Fraud Detection Module
Detects anomalous ticketing behavior using unsupervised learning.
Multi-Objective Scheduling Module
Generates Pareto-optimal schedules using evolutionary algorithms.

Outputs

Demand-aware seat allocation plans
Uncertainty-aware arrival time predictions
Risk-scored inspection recommendations
Optimized train schedules with trade-off visualization
The system directly addresses the gaps identified in Section 6 by integrating predictive analytics, optimization, and explainable AI into a unified framework.

References

[1] J. G. de Gooijer and R. J. Hyndman, "25 years of time series forecasting," *Int. J. Forecast.*, vol. 22, no. 3, pp. 443–473, 2006.
[2] S. Bandara et al., "Passenger demand analysis in Sri Lanka Railways," *Transp. Res. Procedia*, vol. 48, pp. 1234–1245, 2020.
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[5] R. Corman et al., "Railway traffic management in complex networks," *J. Rail Transp. Plan. Manag.*, vol. 4, no. 2, pp. 1–13, 2014.


INPUT DATA SOURCES
Train Schedules & Routes
Passenger Booking Requests
Historical Ticket Sales Data
GPS / IoT Sensors
Passenger Identity Data
INTELLIGENT MODULES
Scheduling & Capacity Optimization
AI Demand Forecasting & Seat Allocation
Train Tracking & Delay Prediction
Fraud Detection & Verification
Advanced Railway Booking and Tracking System
real-time status & tickets
schedules & insights
alerts
OUTPUTS & USERS
Passengers
Railway Administration
Alert & Notification System

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful execution of this project requires expertise across **AI, data analytics, optimization, simulation, and intelligent transportation systems**. To achieve this, team members will systematically acquire specialized domain knowledge through multiple pathways:

Domain Expertise Acquisition:

Detailed study of **Sri Lanka Railway operational manuals, timetable structures, ticketing processes, and historical delay patterns**.

Engagement with railway operations personnel for contextual insights on train dispatching, seat allocation, and fraud inspection workflows.

Analysis of publicly available railway datasets, timetables, and historical passenger flows to establish realistic simulation and forecasting baselines.

Technical Knowledge and Skills Development:

Advanced Machine Learning: Training and deployment of deep learning architectures (LSTM, GRU, Temporal CNN) for spatio-temporal forecasting, sequence prediction, and anomaly detection.

Optimization and Decision Science: Implementation of **multi-objective evolutionary algorithms (NSGA-II)** for scheduling and capacity allocation under uncertainty, including development of Pareto fronts.

Data Engineering: Streaming pipelines for real-time GPS, station, and timetable data ingestion, cleaning, and map-matching.

Simulation & Visualization: Discrete-event simulations to evaluate operational scenarios and interactive dashboards for operator decision support.

Explainable AI (XAI): Integration of attention mechanisms and risk-scoring frameworks for interpretable fraud detection.

Data Acquisition and Preparation:

Historical ticketing records, simulated passenger demand datasets, and GPS/time-stamped train movement data will be anonymized and aggregated to ensure privacy.

Synthetic datasets will be generated for scenarios lacking sufficient historical coverage, including rare fraud events, peak demand periods, and disruption simulations.

Data augmentation techniques will ensure robustness and generalization of predictive models.

Ethical Considerations:

Ethical clearance will be obtained for all research involving passenger-related datasets All personally identifiable information (PII) will be removed to comply with data protection regulations.Risk assessment will be conducted for automated decision modules, ensuring fairness and minimizing biases in passenger-facing applications.

9. Objectives and Novelty

Main Objective

To design and implement an **integrated, AI-driven decision support system** that improves operational efficiency, passenger service quality, and fraud mitigation in Sri Lanka Railways while ensuring fairness and robustness under uncertainty.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22126610 – Muthukudaarachchi V.U	Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification	1. Define a comprehensive railway fraud taxonomy, including ticket reuse, abnormal concession usage, and suspicious resale behaviors, and generate labeled synthetic or anonymized transaction datasets reflecting realistic passenger patterns. 2. Engineer behavioral, temporal, and contextual features from ticketing data and apply unsupervised and semi-supervised anomaly detection techniques such as Isolation Forests, One-Class SVMs, and autoencoder-based models. 3. Develop a risk-scoring and explanation framework that highlights anomaly severity and key contributing features to support real-time inspector decision-making. 4. Evaluate fraud detection performance using precision, recall, F1-score, and ROC-AUC, and	The proposed approach introduces a **proactive, explainable fraud detection framework** for railway ticketing that shifts inspection practices from manual, random checks to **risk-aware, data-driven enforcement**. A distinctive contribution is the explicit analysis of the **trade-off between fraud capture and inspector workload**, an aspect rarely examined in transportation fraud detection studies.

		analyze the trade-off between detection accuracy and inspector workload under varying risk thresholds.	
IT22051202 – Didulantha C S	Intelligent Real-Time Train Tracking and Probabilistic Delay Prediction Framework	1. Design and implement a real-time data ingestion and processing pipeline to stream GPS signals, station timestamps, and timetable data, including data cleaning and map-matching to railway network topology. 2. Engineer delay propagation features capturing temporal dependencies, route congestion, and upstream disruptions, and train regression and sequence-based models (Gradient Boosting, LSTM) for multi-horizon arrival time prediction. 3. Generate probabilistic estimated times of arrival (ETAs) by producing point predictions alongside calibrated confidence intervals to represent uncertainty in delay predictions. 4. Evaluate predictive performance against schedule-only and deterministic baselines using MAE, RMSE, and interval calibration metrics, and assess robustness under simulated disruption and peak-traffic scenarios.	This component advances beyond conventional train tracking systems by delivering **uncertainty-aware, probabilistic delay predictions** tailored to Sri Lankan railway operations. By explicitly modeling delay propagation and prediction uncertainty, the framework enables risk-aware passenger information services and operational decision-making—capabilities largely absent in existing local railway applications and research.

IT22217240 – Narasinghe N.K.B.N.N	Multi-Objective Optimization Framework for Train Scheduling and Capacity Management	1. Formalize railway scheduling and capacity constraints, including track headways, rolling stock availability, crew assignments, and station dwell times, along with conflicting objectives such as delay minimization, overcrowding reduction, utilization maximization, and robustness enhancement. 2. Integrate passenger demand forecasts and delay patterns obtained from Components 1 and 2 into a unified multi-objective scheduling and capacity optimization model. 3. Implement evolutionary and metaheuristic optimization algorithms (e.g., NSGA-II) to generate Pareto-optimal train schedules and coach allocation plans. 4. Develop a discrete-event simulation environment to evaluate candidate schedules under stochastic demand and disruption scenarios, and visualize Pareto fronts and key performance indicators in an operator-oriented dashboard.	This component presents a **research-grade, multi-objective optimization and simulation framework** customized for Sri Lanka Railways. Rather than producing a single heuristic timetable, the framework generates and analyzes **entire sets of Pareto-optimal solutions**, offering transparent insights into trade-offs between passenger service quality, operational efficiency, and robustness under uncertainty—an approach not currently employed in local railway planning.
IT22151506 – D.M.N.T Senawirathna	Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation using AI	1. Collect, integrate, and preprocess multi-source historical and simulated railway ticketing data, calendar information, seasonal events, and route characteristics to construct high-resolution origin–destination (OD) demand matrices. 2. Develop and evaluate baseline statistical forecasting models (ARIMA, Prophet) and advanced deep learning architectures (LSTM, GRU, Temporal Convolutional Networks) to model spatio-temporal passenger demand dynamics.	This component introduces a **Sri Lanka–specific spatio-temporal demand forecasting framework** tightly coupled with an **adaptive, fairness-aware seat allocation mechanism**. Unlike existing rule-based railway systems, the proposed approach explicitly quantifies trade-offs between demand uncertainty, passenger fairness, and seat utilization through predictive modeling and controlled simulation, providing a data-driven foundation for intelligent capacity management.

		3. Formulate an adaptive seat allocation problem incorporating predicted demand, capacity constraints, fairness criteria, and operational policies, and solve it using integer linear programming and heuristic optimization techniques. 4. Conduct large-scale simulation experiments to compare adaptive seat allocation strategies against existing static allocation rules under normal and peak-demand scenarios, using metrics such as MAE, RMSE, MAPE, seat utilization, booking rejection rates, and fairness indices.	

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Data required.

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Chathurangika	Kahadawaarachchi	
Co-Supervisor	Mr	Malidha	Dhammadeeth.	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

Supervisor Signature Unavailable – Explanation

